

Q1. File Replication

File Replication in a distributed File System (DFS) refers to the process of maintaining multiple copies of the same data across different nodes in the system. This ensures data availability, fault tolerance, and reliability, even if one or more nodes fail.

Advantages -

1. **Fault Tolerance** - If a node fails, data can still be accessed from another replica.
2. **High Availability** - Data remains accessible even during maintenance or network failure.
3. **Load Balancing** - Requests can be distributed across replicas, improving performance.
4. **Improved Reliability** - Redundant copies protect against corruption or data loss.

Challenges -

1. **Consistency Maintenance** - Ensuring all replicas remain synchronised.
2. **Storage Overhead** - Multiple copies increase storage requirements.
3. **Network Overhead** - Frequent updates to replicas may cause high bandwidth ~~change~~ usage.
4. **Replica Placement** - Choosing optimal nodes for storing replicas is complex.

Atomic Transactions for Reliability -

Atomic transactions ensure that file operations such as write or update are all-or-nothing. If a write operation fails, changes are rolled back to maintain consistency. This prevents corruption in replicated data and ensures data reliability.

Hadoop Distributed File System

HDFS replicates each file block (default 3 replicas) on different data nodes:

- One replica on the local node
- One on a different rack (for fault tolerance)
- Another on a random node

If a node fails, the NameNode detects it and triggers replication to maintain the replication factor. This ensures continuous reliability and availability of data in large-scale distributed environments.

Q2.

MapReduce is a programming framework in Hadoop used for processing large-scale data in parallel across a distributed cluster.

Working Principle -

1. Input Splitting -

The input dataset is divided into smaller chunks, each processed by a separate map task.

2. Mapping -

Each map task processes the input split and generates intermediate key-value pairs. Example - For word count, map emits (word, 1).

3. Shuffling and Sorting -

The framework groups all values with the same key together and sorts them.

4. Reducing

Reducers aggregate the values for each key to produce the final output.

Example - Sum all counts for each word
→ (word, total-count)

5. Output

The reducer writes the final results to HDFS.

- maple - counting words in documents:
- Map Output: (word, 1)
 - Reduce output: (word, total occurrences)
- The model ensures parallel processing, fault tolerance and scalability.

Q3.

The Seven-Step Cloud Migration Model provides a structured approach for moving IT infrastructure, applications or data to a cloud environment.

STEP	DESCRIPTION
1. Assessment	Evaluate current infrastructure, dependencies and readiness for cloud migration.
2. Planning	Define objectives, choose suitable cloud models (public/private/hybrid), and set timelines.
3. Proof of Concept	Test migration with a small workload to validate strategy and performance.
4. Data Migration	Transfer databases, files, and other assets securely to the cloud.
5. Application Migration	Move or reconfigure applications to operate in the cloud environment.
6. Optimization	Fine tune performance, scalability and cost using cloud-native tools.
7. Monitoring and Management	Continuously monitor resource utilization and security after migration.

This model ensures minimal disruption, better risk management, and efficient adoption of cloud technologies.

Q4. layers of Cloud Computing -

1. Infrastructure Layer - Physical hardware and network resources.
2. Platform Layer - Runtime Environment and development tools.
3. Application Layer - End-user applications delivered over the internet.

Types of Cloud Computing -

1. Public Cloud - Services offered over the internet (AWS, Azure)
2. Private Cloud - Dedicated infrastructure for one organization
3. Hybrid Cloud - Combination of public and private clouds
4. Community Cloud - Shared by organizations with common goals (Education, healthcare).

Cloud Service Models -

Model	Description	Real-World Examples
IaaS (Infrastructure as a Service)	Provides a virtualized computing resources like VMs, storage and networking	AWS, EC2, Google Computer Engine
PaaS (Platform as a Service)	Provides development platforms, tools and environments for building applications	Google App Engine, Microsoft Azure App Service
SaaS (Software as a Service)	Provides ready-to-use applications hosted on the cloud.	Google Workplace, Salesforce, Zoom

SUMMARY -

- IaaS - Manage apps, providers manages infrastructure
- PaaS - Manage code, provider manages platform
- SaaS - Use software, provider manages everything.